

AP CALCULUS AB - UNIT 1

Guided Notes 1.6: Determining Limits Using Algebraic Manipulation



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we use substitution as a diagnostic rather than a final answer when it produces an indeterminate form?

- Use substitution as a diagnostic rather than a final answer when it produces an indeterminate form.
- Select factoring, conjugates, common denominators, trigonometric identities, or side-specific algebra efficiently.
- Justify cancellation through equality on a punctured neighborhood.
- Use the standard trigonometric limits in radians.

Key Knowledge, Vocabulary, and Notation

Diagnostic substitution

A finite value completes the problem; $0/0$ signals that algebra may reveal hidden local behavior.

Punctured equivalence

Expressions that agree for all nearby $x \neq c$ have the same limit at c .

Factoring

Cancel common multiplicative factors, never terms separated by addition.

Conjugates

Use $(A - B)(A + B) = A^2 - B^2$ to expose a factor.

Trig anchors

$\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$ and $\lim_{u \rightarrow 0} \frac{1 - \cos u}{u} = 0$ in radians.

Concept Development and Strategy

1. First substitute and classify the result. Then choose the shortest structure-preserving rewrite.
2. State the excluded target when canceling a factor; the simplified expression is used only for nearby inputs.
3. For absolute value and piecewise expressions, simplify separately on each side before comparing.

Worked Example: Factoring a removable form

Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

Answer and Reasoning

Factor: $\frac{(x-5)(x+5)}{x-5} = x + 5$ for $x \neq 5$. Therefore the limit is 10.

Worked Example: Rationalizing a numerator

Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x+5} - 3}{x - 4}$.

Answer and Reasoning

Multiply by the conjugate. The expression becomes $\frac{1}{\sqrt{x+5}+3}$ for $x \neq 4$, so the limit is $1/6$.

Worked Example: Trigonometric scaling

Evaluate $\lim_{x \rightarrow 0} \frac{\sin(7x)}{x}$.

Answer and Reasoning

Write $7 \frac{\sin(7x)}{7x}$. Since $7x \rightarrow 0$, the limit is 7.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

2. Evaluate $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2}$.

3. Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.

4. Evaluate $\lim_{x \rightarrow 4} \frac{\frac{1}{x} - \frac{1}{4}}{x - 4}$.

5. Evaluate $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2}$.

6. Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{\sqrt{x + 2} - 2}$.

7. Prove that for positive integer n , $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ equals

(A) 0

(B) 3

(C) 6

(D) DNE

2. $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$ equals

(A) $\frac{3}{4}$

(B) $\frac{4}{3}$

(C) 3

(D) 4

3. A calculator suggests $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} \approx 2$. Which table best supports the conclusion?

(A) Only $x = 0$

(B) Inputs near 0 from both sides with increasing precision

(C) Only positive integers

(D) Only $x = 1, 2, 3$

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+3x} - \sqrt{1-2x}}{x}$.

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